PBL Portal Content - Hansraj Pandey

Smart Hospital Management System (SmartCare) | Team Nobita (PBL2627-AI&DS-A-031;) | Mentor: Er. Ram Babu Buri

Member	Hansraj Pandey (24E1ARADM40P055)
Module	Module 3 - Appointment and Consultation
Code owned	Appointment model, AppointmentDAO, AppointmentService (double-booking check), AppointmentServlet, MedicalRecord model and DAO, patient book_appointment/appointments JSPs, receptionist book_appointment/appointments JSPs, doctor appointments and consultation JSPs.
Period	06-07-2026 to 04-10-2026 (13 weeks). Daily hours between 1.5 and 2.0.
GitHub link	https://github.com/mathurdevansh/Smart-Hospital-Management-System

How to use this file

1) Daily Work Log tab: pick the Date, type Hours Worked, keep Status as shown, paste the description, click Submit Today's Work. 2) Weekly Report tab: choose the Week, paste the GitHub link, then paste the three boxes (Work Completed, Problems Faced, Plan for Next Week) from Part A. 3) Part A (weekly) comes first, Part B has the day-wise table. Submit your own entries only, from your own login.

PART A - Weekly Reports (13 weeks)

Week 1 | 06-07-2026 to 12-07-2026 | Phase: Team formation, guide selection and abstract

Week dropdown	Week 1
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Joined team of five and listed hospital workflows to digitise. (2) Studied how appointment booking is handled in clinics. (3) Wrote the abstract section on appointments and consultation. (4) Discussed the technology stack and guide expectations. (5) Proofread the abstract and supported submission.
Problems Faced	Choosing the right scope for the idea. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Listed functional requirements for booking, rescheduling and cancelling. (2) Wrote requirements for doctor consultation and diagnosis notes. (3) Defined appointment statuses: Scheduled, Completed and Cancelled.

Week 2 | 13-07-2026 to 19-07-2026 | Phase: SRS (Software Requirement Specification)

Week dropdown	Week 2
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Listed functional requirements for booking, rescheduling and cancelling. (2) Wrote requirements for doctor consultation and diagnosis notes. (3) Defined appointment statuses: Scheduled, Completed and Cancelled. (4) Added business rules such as no double booking of a doctor slot. (5) Reviewed the appointment part of the SRS.
Problems Faced	Defining all appointment statuses clearly. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Drew the use-case diagram for booking and consultation. (2) Designed the class diagram for Appointment. (3) Designed the class diagram for MedicalRecord.

Week 3 | 20-07-2026 to 26-07-2026 | Phase: UML design (use-case, class, sequence)

Week dropdown	Week 3
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Drew the use-case diagram for booking and consultation. (2) Designed the class diagram for Appointment. (3) Designed the class diagram for MedicalRecord. (4) Drew the sequence diagram for booking an appointment. (5) Merged module UML into the common design document.
Problems Faced	Showing the double-booking rule in sequence diagram. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Designed the appointments table with doctor, patient and slot columns. (2) Designed the medical_records table linked to appointments. (3) Worked out indexes for date and doctor lookups.

Week 4 | 27-07-2026 to 02-08-2026 | Phase: Database (ER) design, UI mock-ups, project setup

Week dropdown	Week 4
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Designed the appointments table with doctor, patient and slot columns. (2) Designed the medical_records table linked to appointments. (3) Worked out indexes for date and doctor lookups. (4) Cloned the repo, set up Tomcat and tested the sample page. (5) Drew mock-ups for the booking form and consultation screen.
Problems Faced	Choosing correct indexes for lookups. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote the Appointment model with date and time fields. (2) Wrote the MedicalRecord model class. (3) Implemented AppointmentDAO insert and fetch by patient.

Week 5 | 03-08-2026 to 09-08-2026 | Phase: Module coding - models and DAO layer

Week dropdown	Week 5
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Wrote the Appointment model with date and time fields. (2) Wrote the MedicalRecord model class. (3) Implemented AppointmentDAO insert and fetch by patient. (4) Added AppointmentDAO fetch by doctor and by date. (5) Implemented isSlotBooked query and MedicalRecordDAO methods.
Problems Faced	SQL Time type mapping was confusing. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Created AppointmentService with input validation. (2) Added the double-booking check inside the service. (3) Implemented AppointmentServlet for booking requests.

Week 6 | 10-08-2026 to 16-08-2026 | Phase: Module coding - services and servlets

Week dropdown	Week 6
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Created AppointmentService with input validation. (2) Added the double-booking check inside the service. (3) Implemented AppointmentServlet for booking requests. (4) Added reschedule and cancel actions in the servlet. (5) Linked status updates and audit logging for each action.
Problems Faced	Double-booking check allowed edge cases. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Created the patient book_appointment.jsp with doctor and slot selection. (2) Created the patient appointments.jsp with status badges. (3) Created the receptionist book_appointment.jsp.

Week 7 | 17-08-2026 to 23-08-2026 | Phase: Module coding - JSP views and UI

Week dropdown	Week 7
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Created the patient book_appointment.jsp with doctor and slot selection. (2) Created the patient appointments.jsp with status badges. (3) Created the receptionist book_appointment.jsp. (4) Created the doctor appointments.jsp for the daily queue. (5) Created the doctor consultation.jsp to record diagnosis.
Problems Faced	Date picker accepted past dates. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Added past-date and empty reason validations. (2) Handled booking exceptions with friendly messages. (3) Added JavaScript to block past dates in the date picker.

Week 8 | 24-08-2026 to 30-08-2026 | Phase: Module coding - validation, exceptions, security

Week dropdown	Week 8
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Added past-date and empty reason validations. (2) Handled booking exceptions with friendly messages. (3) Added JavaScript to block past dates in the date picker. (4) Restricted cancel action to the owner or receptionist. (5) Tested validation messages across forms.
Problems Faced	Cancel was visible to wrong role. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Completed consultation saving into medical records. (2) Added the status change to Completed after consultation. (3) Added filters on appointment lists by date and status.

Week 9 | 31-08-2026 to 06-09-2026 | Phase: Module coding - module completion and self-review

Week dropdown	Week 9
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Completed consultation saving into medical records. (2) Added the status change to Completed after consultation. (3) Added filters on appointment lists by date and status. (4) Reviewed code and added comments in service and DAO. (5) Pushed module code to GitHub and updated the sheet.
Problems Faced	Status did not update after consultation. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Integrated doctor list from Module 2 into booking. (2) Connected consultation with prescription and lab pages. (3) Fixed time format mismatch between JSP and DAO.

Week 10 | 07-09-2026 to 13-09-2026 | Phase: Integration of all five modules

Week dropdown	Week 10
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Integrated doctor list from Module 2 into booking. (2) Connected consultation with prescription and lab pages. (3) Fixed time format mismatch between JSP and DAO. (4) Verified patient, receptionist and doctor views on one appointment. (5) Ran full flow from booking to consultation.
Problems Faced	Time format mismatch between JSP and DAO. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote test cases for booking, rescheduling and cancelling. (2) Tested double-booking prevention with same slot. (3) Tested consultation notes and record saving.

Week 11 | 14-09-2026 to 20-09-2026 | Phase: Testing (unit, functional, role-based)

Week dropdown	Week 11
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Wrote test cases for booking, rescheduling and cancelling. (2) Tested double-booking prevention with same slot. (3) Tested consultation notes and record saving. (4) Tested role restrictions on appointment pages. (5) Fixed issues from testing and retested.
Problems Faced	A test case failed for same-time reschedule. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Tested booking flow on the live Render site. (2) Fixed time zone difference between local and cloud server. (3) Corrected status display bug on the appointments list.

Week 12 | 21-09-2026 to 27-09-2026 | Phase: Deployment (Docker, Render, cloud MySQL) and bug fixing

Week dropdown	Week 12
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Tested booking flow on the live Render site. (2) Fixed time zone difference between local and cloud server. (3) Corrected status display bug on the appointments list. (4) Retested the consultation page on the live site. (5) Updated notes for deployment problems and fixes.
Problems Faced	Time zone differed on cloud server. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote the appointment module section of the report. (2) Added flow diagrams and screenshots to the report. (3) Prepared slides on the appointment workflow.

Week 13 | 28-09-2026 to 04-10-2026 | Phase: Report, PPT, demo video and viva preparation

Week dropdown	Week 13
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Wrote the appointment module section of the report. (2) Added flow diagrams and screenshots to the report. (3) Prepared slides on the appointment workflow. (4) Rehearsed the booking and consultation demo. (5) Prepared viva answers on service layer and slot checks.
Problems Faced	Exporting screenshots for report. Solved with team discussion and guide's suggestions.
Plan for Next Week	Final viva and demo, submit report, PPT, SRS, demo video and working software on the portal.

PART B - Daily Work Log (Monday to Friday, 65 entries)

Status for all entries: Completed. Saturday is the weekly mentor meeting, so no daily entry is listed.

Week 1 - Team formation, guide selection and abstract

Date (DD-MM-YYYY)	Hours	Status	Work Description
06-07-2026 (Mon)	1.75	Completed	Joined team of five and listed hospital workflows to digitise.
07-07-2026 (Tue)	2	Completed	Studied how appointment booking is handled in clinics.
08-07-2026 (Wed)	2	Completed	Wrote the abstract section on appointments and consultation.
09-07-2026 (Thu)	1.5	Completed	Discussed the technology stack and guide expectations.
10-07-2026 (Fri)	1.75	Completed	Proofread the abstract and supported submission.

Week 2 - SRS (Software Requirement Specification)

Date (DD-MM-YYYY)	Hours	Status	Work Description
13-07-2026 (Mon)	1.75	Completed	Listed functional requirements for booking, rescheduling and cancelling.
14-07-2026 (Tue)	2	Completed	Wrote requirements for doctor consultation and diagnosis notes.
15-07-2026 (Wed)	1.5	Completed	Defined appointment statuses: Scheduled, Completed and Cancelled.
16-07-2026 (Thu)	1.5	Completed	Added business rules such as no double booking of a doctor slot.
17-07-2026 (Fri)	2	Completed	Reviewed the appointment part of the SRS.

Week 3 - UML design (use-case, class, sequence)

Date (DD-MM-YYYY)	Hours	Status	Work Description
20-07-2026 (Mon)	1.5	Completed	Drew the use-case diagram for booking and consultation.
21-07-2026 (Tue)	2	Completed	Designed the class diagram for Appointment.
22-07-2026 (Wed)	2	Completed	Designed the class diagram for MedicalRecord.
23-07-2026 (Thu)	1.75	Completed	Drew the sequence diagram for booking an appointment.
24-07-2026 (Fri)	1.5	Completed	Merged module UML into the common design document.

Week 4 - Database (ER) design, UI mock-ups, project setup

Date (DD-MM-YYYY)	Hours	Status	Work Description
27-07-2026 (Mon)	2	Completed	Designed the appointments table with doctor, patient and slot columns.
28-07-2026 (Tue)	1.75	Completed	Designed the medical_records table linked to appointments.
29-07-2026 (Wed)	1.5	Completed	Worked out indexes for date and doctor lookups.
30-07-2026 (Thu)	1.75	Completed	Cloned the repo, set up Tomcat and tested the sample page.
31-07-2026 (Fri)	1.75	Completed	Drew mock-ups for the booking form and consultation screen.

Week 5 - Module coding - models and DAO layer

Date (DD-MM-YYYY)	Hours	Status	Work Description
03-08-2026 (Mon)	2	Completed	Wrote the Appointment model with date and time fields.
04-08-2026 (Tue)	1.5	Completed	Wrote the MedicalRecord model class.
05-08-2026 (Wed)	1.75	Completed	Implemented AppointmentDAO insert and fetch by patient.
06-08-2026 (Thu)	2	Completed	Added AppointmentDAO fetch by doctor and by date.
07-08-2026 (Fri)	1.5	Completed	Implemented isSlotBooked query and MedicalRecordDAO methods.

Week 6 - Module coding - services and servlets

Date (DD-MM-YYYY)	Hours	Status	Work Description
10-08-2026 (Mon)	2	Completed	Created AppointmentService with input validation.
11-08-2026 (Tue)	1.5	Completed	Added the double-booking check inside the service.
12-08-2026 (Wed)	1.75	Completed	Implemented AppointmentServlet for booking requests.
13-08-2026 (Thu)	2	Completed	Added reschedule and cancel actions in the servlet.
14-08-2026 (Fri)	2	Completed	Linked status updates and audit logging for each action.

Week 7 - Module coding - JSP views and UI

Date (DD-MM-YYYY)	Hours	Status	Work Description
17-08-2026 (Mon)	2	Completed	Created the patient book_appointment.jsp with doctor and slot selection.
18-08-2026 (Tue)	1.5	Completed	Created the patient appointments.jsp with status badges.
19-08-2026 (Wed)	2	Completed	Created the receptionist book_appointment.jsp.
20-08-2026 (Thu)	2	Completed	Created the doctor appointments.jsp for the daily queue.
21-08-2026 (Fri)	1.75	Completed	Created the doctor consultation.jsp to record diagnosis.

Week 8 - Module coding - validation, exceptions, security

Date (DD-MM-YYYY)	Hours	Status	Work Description
24-08-2026 (Mon)	2	Completed	Added past-date and empty reason validations.
25-08-2026 (Tue)	1.75	Completed	Handled booking exceptions with friendly messages.
26-08-2026 (Wed)	2	Completed	Added JavaScript to block past dates in the date picker.
27-08-2026 (Thu)	2	Completed	Restricted cancel action to the owner or receptionist.
28-08-2026 (Fri)	1.5	Completed	Tested validation messages across forms.

Week 9 - Module coding - module completion and self-review

Date (DD-MM-YYYY)	Hours	Status	Work Description
31-08-2026 (Mon)	1.75	Completed	Completed consultation saving into medical records.
01-09-2026 (Tue)	1.5	Completed	Added the status change to Completed after consultation.
02-09-2026 (Wed)	1.5	Completed	Added filters on appointment lists by date and status.
03-09-2026 (Thu)	1.75	Completed	Reviewed code and added comments in service and DAO.
04-09-2026 (Fri)	1.75	Completed	Pushed module code to GitHub and updated the sheet.

Week 10 - Integration of all five modules

Date (DD-MM-YYYY)	Hours	Status	Work Description
07-09-2026 (Mon)	1.5	Completed	Integrated doctor list from Module 2 into booking.
08-09-2026 (Tue)	2	Completed	Connected consultation with prescription and lab pages.
09-09-2026 (Wed)	1.75	Completed	Fixed time format mismatch between JSP and DAO.
10-09-2026 (Thu)	1.75	Completed	Verified patient, receptionist and doctor views on one appointment.
11-09-2026 (Fri)	2	Completed	Ran full flow from booking to consultation.

Week 11 - Testing (unit, functional, role-based)

Date (DD-MM-YYYY)	Hours	Status	Work Description
14-09-2026 (Mon)	2	Completed	Wrote test cases for booking, rescheduling and cancelling.
15-09-2026 (Tue)	1.75	Completed	Tested double-booking prevention with same slot.
16-09-2026 (Wed)	1.75	Completed	Tested consultation notes and record saving.
17-09-2026 (Thu)	2	Completed	Tested role restrictions on appointment pages.
18-09-2026 (Fri)	1.75	Completed	Fixed issues from testing and retested.

Week 12 - Deployment (Docker, Render, cloud MySQL) and bug fixing

Date (DD-MM-YYYY)	Hours	Status	Work Description
21-09-2026 (Mon)	1.75	Completed	Tested booking flow on the live Render site.
22-09-2026 (Tue)	1.5	Completed	Fixed time zone difference between local and cloud server.
23-09-2026 (Wed)	1.75	Completed	Corrected status display bug on the appointments list.
24-09-2026 (Thu)	2	Completed	Retested the consultation page on the live site.
25-09-2026 (Fri)	1.75	Completed	Updated notes for deployment problems and fixes.

Week 13 - Report, PPT, demo video and viva preparation

Date (DD-MM-YYYY)	Hours	Status	Work Description
28-09-2026 (Mon)	1.5	Completed	Wrote the appointment module section of the report.
29-09-2026 (Tue)	2	Completed	Added flow diagrams and screenshots to the report.
30-09-2026 (Wed)	2	Completed	Prepared slides on the appointment workflow.
01-10-2026 (Thu)	1.75	Completed	Rehearsed the booking and consultation demo.
02-10-2026 (Fri)	2	Completed	Prepared viva answers on service layer and slot checks.

Total hours in this log: 116.25 over 65 days.